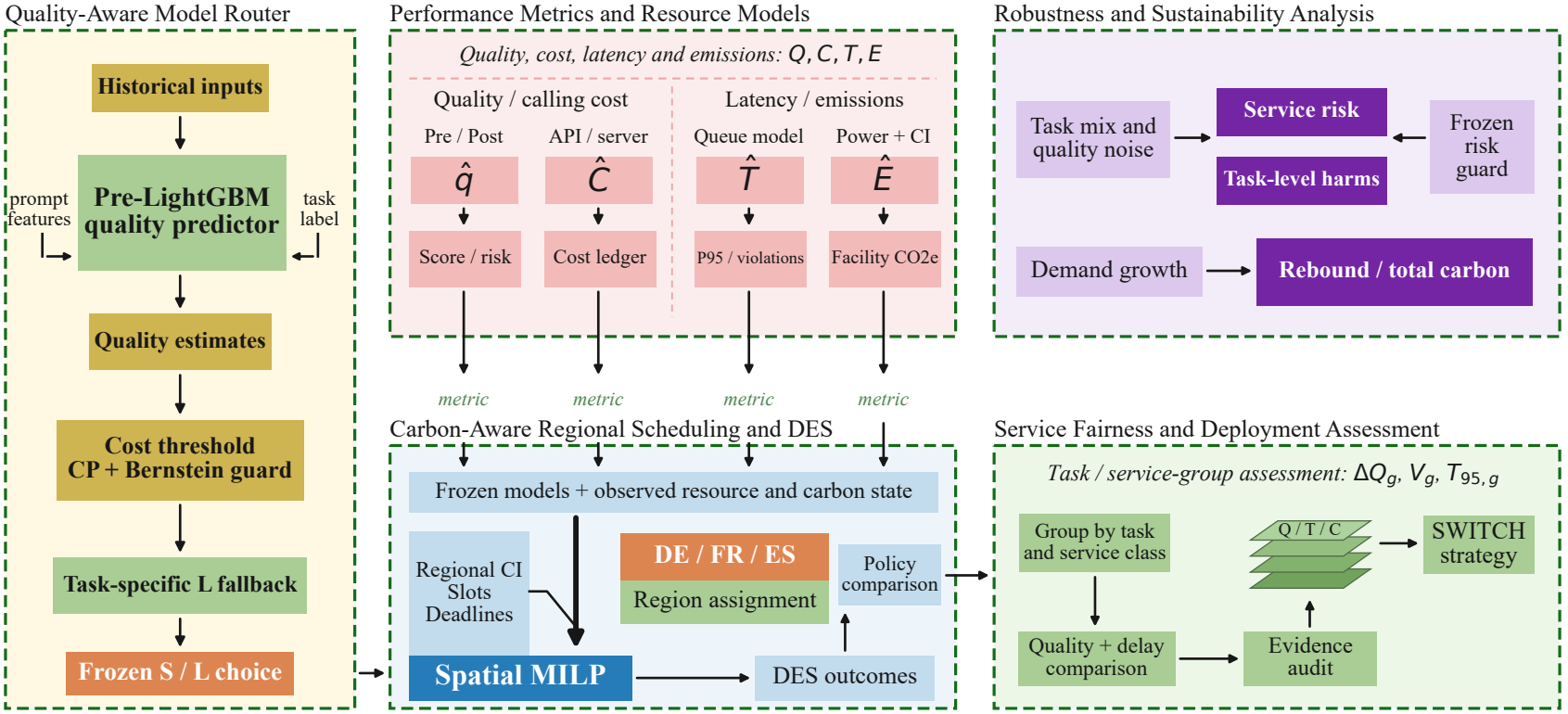

Quality-Aware Model Router
Historical inputs
prompt features
Pre-LightGBM quality predictor
task label
Quality estimates
Cost threshold CP + Bernstein guard
Task-specific L fallback
Frozen S / L choice
Performance Metrics and Resource Models
Quality, cost, latency and emissions: Q, C, T, E
Quality / calling cost
Latency / emissions
Pre / Post
API / server
Queue model
Power + CI
$\hat{q}$
$\hat{C}$
$\hat{T}$
$\hat{E}$
Score / risk
Cost ledger
P95 / violations
Facility CO2e
metric
metric
metric
metric
Robustness and Sustainability Analysis
Task mix and quality noise
Service risk
Task-level harms
Frozen risk guard
Demand growth
Rebound / total carbon
Carbon-Aware Regional Scheduling and DES
Frozen models + observed resource and carbon state
Regional CI Slots Deadlines
DE / FR / ES
Region assignment
Policy comparison
Spatial MILP
DES outcomes
Service Fairness and Deployment Assessment
Task / service-group assessment: ΔQ_g, V_g, $T_{95,g}$
Group by task and service class
Q / T / C
SWITCH strategy
Quality + delay comparison
Evidence audit
Our work